Deftaudio PCM Custom Card for Korg DDD-1/DDD-5

Technical Description

The Deftaudio PCM Custom Card is a programmable PCM expansion module compatible with the Korg DDD-1, DDD-5, and Korg DRM series drum machines. The card provides eight independent PCM banks stored in non-volatile memory, selectable via an onboard DIP switch interface. It is based on the open-source PicoROM architecture and supports user modification at both hardware and software levels.

Specifications:

1. Compatibility (Korg PCM card interface–compatible models)
    - Korg DDD-1
    - Korg DDD-5
    - Korg DRM

2. PCM Banks and Memory Structure
    - 8 selectable PCM banks, stored in flash-based non-volatile memory.
    - Bank selection performed via a front-mounted 8-position DIP switch.
    - Each bank contains up to 8 PCM sound slots.
    - Maximum available sample time per bank: ~1.5 seconds, shared across all sound slots.
    - Output format: 8-bit PCM samples compatible with Korg DDD/DRM playback requirements.

3. Host Integration
    - PCM sound names programmed on the card are correctly recognized and displayed on Korg DDD-1 and DDD-5 front-panel screens.
    - Fully compatible with factory PCM card timing, addressing, and data lines.

4. Programming Interface
    - USB-C port for loading PCM sample data and metadata.
    - Programming tools and scripts included for Windows.
    - macOS and Linux users may run the utilities via Parallels, VirtualBox, Wine, or equivalent environments.
    - Supports standard WAV files with no special requirements. Sounds are automatically trimmed, resampled, mono converted to maximize utilization of internal memory.

5. Status Indication
    - Perimeter LED activity indicators provide visual feedback during memory access and during USB programming operations.

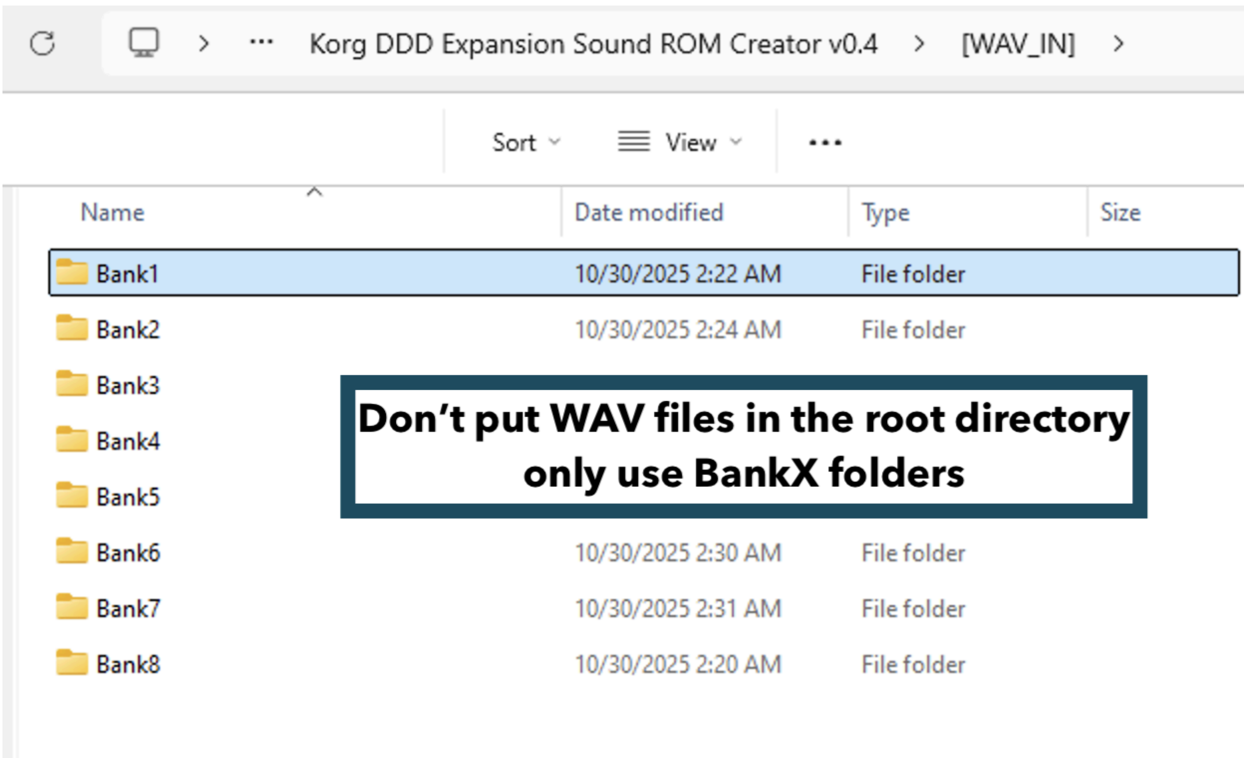
6. Architecture and Modifiability
    - Hardware is based on the PicoROM open-source project.
    - PCM conversion scripts are highly modifiable and tweakable. Based on previous work of R-Massive and shared back with a community.
    - Fully Github documented, the DIY community is welcomed.

Usage:

For basic information, please, refer to the Video Tutorial:
https://youtu.be/BM5A2fx6TgM

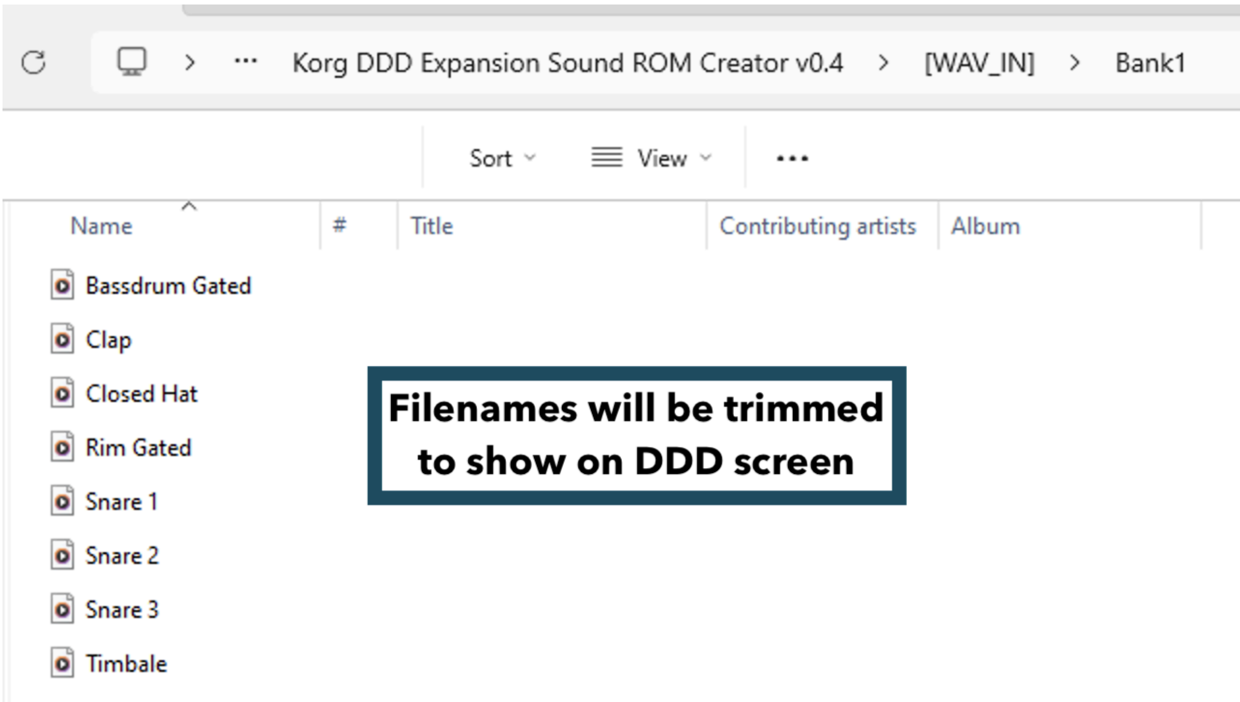
Download programming tools, unpack files:
https://github.com/Deftaudio/

Put your samples under [WAV-IN] in BankX subdirectories for corresponding Banks 1-8.

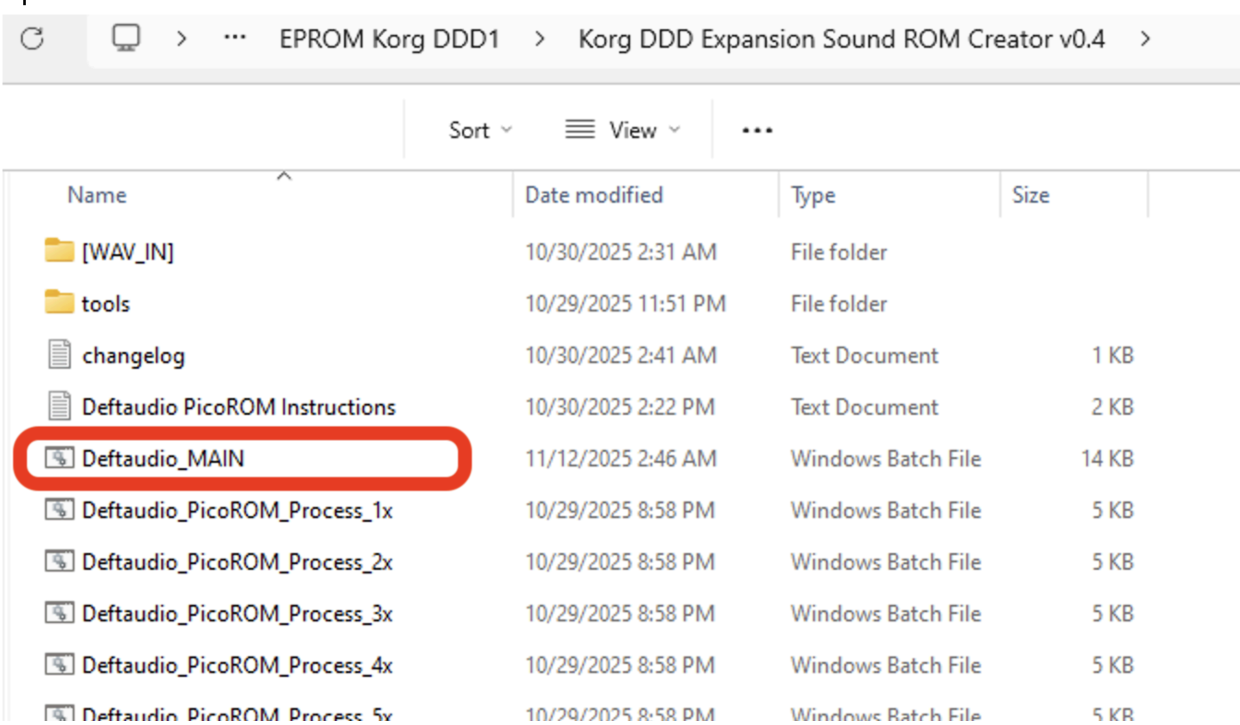


Supports any WAV file format. Scripts will take care of processing.

- Each bank can have between 1 to 8 sounds. The layout will be automatically adjusted and input sounds trimmed to maximize usage of bank memory.
- Sound names are preserved. Eight characters of the original file name will be shown on Korg DDD screen. Avoid using custom characters.
- Sounds will be combined in banks while sorted in alphabetic order.



Double click on *Deftaudio_main.bat* script.
First it scans all subdirectories for files. It prints the total number of files in each directory before it proceeds.



Press any key to start conversion. It takes 10-20 seconds to create the bank
At the end of the script the bank image is created: PicoROM.BIN

The script can program it automatically onto a connected card. If you have multiple cards, only one card at a time needs to be connected. Each PCM card should have a “PicoROM” name programmed for the script to recognize it (all assembled units have it already).

Manual Card programming using PicoROM tools:

Refer to the PicoROM manual:

PicoROM programming tools available for Windows, MacOS, Linux. The syntax is the same across all versions.

Identify PicoROM first:
Picorom-x86_64-pc-windows-msvc list

Program PicoROM.BIN image from the same directory where tools are:
picorom-x86_64-pc-windows-msvc.exe upload -s PicoROM PicoROM.BIN 2mbit

(Optional) PicoROM name change:
Find it first: *Picorom-x86_64-pc-windows-msvc list*
Picorom-x86_64-pc-windows-msvc rename <CURRENT_NAME_OR_ID> <NEW_NAME>

Troubleshooting

Some units may fail to detect the emulator reliably or display a “Card Error” during boot. This is a complex issue with multiple possible causes; oxidized card slots and marginal power delivery are the most common.
Any looseness in the card slot can result in poor contact. Additionally, some slots have simply seen heavy use over the past 30 years and may be mechanically worn or weak.
If you encounter these issues, try installing the original card (if available). This helps distinguish mechanical or contact problems from power-related issues. Original cards draw less current and may operate correctly even when the emulator pushes the power subsystem to its limits. Some DDD-1 / DDD-5 units cannot provide stable slot power due to aging capacitors and may require recapping.

Before opening the drum machine, you can try the following workarounds:

- Start the machine with Bank 0 selected (all DIP switches down). This can improve detection, as no pull-up resistors are engaged.
- Keep the USB-C cable connected to a PC or external power supply. In this configuration, the emulator sources power from USB-C, reducing the load on the internal power supply.